INCH-POUND
MIL-DTL-717G
10 April 2018
SUPERSEDING
MIL-C-717F
12 November 1987
MIL-C-19794B
11 December 1962
(See 6.4)

DETAIL SPECIFICATION

CASTABLE MIX, REFRACTORY, HIGH-TEMPERATURE AND INSULATING, HYDRAULIC SETTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers conventional hydraulic setting high-temperature castable refractory mix for use where temperatures up to 3000 degrees Fahrenheit (°F) may be encountered and hydraulic setting thermal insulating high-temperature castable refractory mixes for use as back up insulation for refractory linings of boilers.

1.2 Classification. Refractory castable mixes are of the following types, as specified (see 6.2).

a. Type I – 3000 °F service temperature conventional castable mix. A dense castable mix of this type falls under ASTM C401, classes E or F.

b. Type II – 3000 °F service temperature insulating castable mix.

c. Type III – 2000 °F service temperature insulating castable mix.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspections by Attributes

(Copies of this document are available online at www.webstore.ansi.org.)

ASTM INTERNATIONAL

ASTM C133 - Standard Test Methods for Cold Crushing Strength and Modulus of Rupture of Refractories

ASTM C134 - Standard Test Methods for Size, Dimensional Measurements, and Bulk Density of Refractory Brick and Insulating Firebrick

ASTM C210 - Standard Test Method for Reheat Change of Insulating Firebrick

ASTM C860 - Standard Test Method for Determining the Consistency of Refractory Castable Using the Ball-In-Hand Test

ASTM C862 - Standard Practice for Preparing Refractory Concrete Specimens by Casting

ASTM C865 - Standard Practice for Firing Refractory Concrete Specimens

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. The material shall be a mixture of Alumina and Alumina-silicates and a hydraulic bond material that mixes readily with potable water. Types I and II castable mix shall contain polymeric fibers to enhance curing. When such de-watering agents are not a standard ingredient in commercial products of the brand name being offered, they shall be added (see 3.9).

3.3 Appearance of castable mix. The castable mix shall be consistent with the type specified (see 1.2 and 6.2). There shall be no foreign, lumped, or caked material not readily broken by hand. Maximum particle size shall be consistent with manufacturer's product data for the brand being supplied. Particle size is not a specification requirement. The intent of this visual particle inspection is to verify that the correct type of castable specified (see 1.2) has been supplied.

3.4 Time for hardening. The material shall remain in a castable consistency for at least 30 minutes after the first addition of mixing water. The hardening time shall be not greater than 24 hours at indoor ambient conditions. At the end of this period, the material shall support its own weight and keep its originally-cast shape (see 4.6.1).

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3.5 Modulus of rupture. The modulus of rupture shall be not less than 500 pounds per square inch (lb/in^2) after drying, and not less than 300 lb/in^2 after being heat soaked at 1500 °F for 5 hours for type I castable mix (see 4.6.2). The modulus of rupture shall be not less than 350 lb/in^2 after drying, and not less than 100 lb/in^2 after being heat soaked at 1500 °F for 5 hours for type II castable mix (see 4.6.2).

3.6 Permanent linear change. The permanent linear change shall not exceed 1.5 percent when heated at 2550 °F for type I (see 4.6.3). The permanent linear change shall not exceed 1.5 percent when heated at 2000 °F for types II and III (see 4.6.3).

3.7 Bulk density. The bulk density of the three types shall not exceed the values shown in table I (see 4.6.4).

TABLE I. Bulk density requirements.

Type	Density	Heat soak temperature and duration
I	Not to exceed 150 lb/ft^3	2550 °F for 5 hours
II	Not to exceed 95 lb/ft^3	2000 °F for 5 hours
III	Not to exceed 45 lb/ft^3	2000 °F for 5 hours

3.8 Shelf-life. The material, when stored at ambient conditions in a container designed to extend shelf-life (see 6.11, level A and B) for a minimum period of 15 months, shall readily mix with potable water in accordance with manufacturer's instructions and meet the requirements as specified in 3.4, 3.5, 3.6, and 3.7 (see 6.5.1).

3.9 Drying additives. Type I and II castable mix shall contain polymeric fibers designed to increase permeability and reduce risk of explosive spalling during initial firing. When polymeric fibers are not included in the brand being offered, the added fibers shall be Polypropylene or Polyethylene of a diameter and length to allow fast dispersion and that is intended for general refractory use. Refer to intended use (see 6.1) for guidance. The fibers shall be factory mixed and integral to the unit packaged material.

3.10 Toxicity and prohibited materials.

3.10.1 Toxicity. When evaluated in accordance with 4.7.1, the castable mix shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.7.1 and 6.6).

3.10.2 Prohibited materials. The castable mix shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2 and 6.3).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of the tests shown in table II. Materials conforming to the requirements specified in table II shall be conditional pending verification of the shelf-life requirement of 3.8.

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TABLE II. First article inspection.

Inspection	Test paragraph	Requirement paragraph
Inclusion of drying agent	4.6.5	3.2, 3.9
Time for hardening	4.6.1	3.4
Modulus of rupture	4.6.2	3.5
Permanent linear change	4.6.3	3.6
Bulk density	4.6.4	3.7
Provide data to NMCPHC (when required)	4.7	3.10

4.3 Conformance inspection. Conformance inspection shall consist of the examinations and tests shown in table III.

TABLE III. Conformance inspection.

Inspection	Test paragraph	Requirement paragraph
Visual examination	4.5	3.3
Time for hardening	4.6.1	3.4
Modulus of rupture	4.6.2	3.5
Permanent linear change	4.6.3	3.6
Bulk density	4.6.4	3.7

4.4 Sample size. Sampling for first article and conformance inspection shall be performed in accordance with ANSI/ASQ Z1.4 and table IV. For purposes of sampling, an inspection lot for examinations and tests shall consist of all material of the same type offered for delivery at one time. No less than five sample units shall be randomly selected throughout the lot. No more than one sample shall be drawn from any one container. Unless otherwise specified (see 6.2), the unit of sample shall be 40 pounds for type I and II and 15 pounds for type III. The sample used for testing shall consist of a 200-pound composite of the unit samples for type I and a 75-pound composite for types II and III. The number of determinations per composite mix is shown in table V.

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TABLE IV. Sampling for first article and conformance inspection.

Inspection	Type	Test paragraph	Inspection level
Visual examination	I, II, and III	4.5	S-1
Time for hardening	I, II, and III	4.6.1	L-2
Modulus of rupture	I and II	4.6.2	1/
Permanent linear change	I, II, and III	4.6.3	1/
Bulk density	I, II, and III	4.6.4	1/
NOTE: 1/ Sample size and accept or reject criteria are as specified in the test paragraph.			

TABLE V. Instructions for testing.

Characteristic	Requirement	Test method	Number of determinations per composite	Results recorded numerically to nearest
Modulus of rupture: Cured at ambient (70 to 90 °F) temperature for 24 hours	3.5	4.6.2.a	Average of 4	lb/in^2
Modulus of rupture: Cured, oven dried and heat soaked at 1500 °F, 5 hours	3.5	4.6.2.b	Average of 4	lb/in^2
Permanent linear change	3.6	4.6.3	Average of 4	0.1 percent
Bulk density	3.7	4.6.4	Average of 3	0.1 lb/ft^3
Time for hardening	3.4	4.6.1	3	Observation recorded at 24 hours; pass or fail conclusion made on each sample
NOTE: All test values upon which results or conclusions are based shall be recorded. Test values for all specimens tested for each characteristic shall be considered when calculating the average results. If a failure occurs, a description of the failure shall be recorded and photographed (optional). Results shall be provided, as specified (see 6.2).				

4.4.1 Sampling for examination of appearance. A minimum of five samples shall be selected from each lot for the examination of 4.5.

4.5 Visual examination. The samples drawn for testing shall be visually examined to ensure conformance with 3.3. The intent of this visual particle inspection is to verify that the correct type of castable mix under 1.2 has been supplied. If the sample fails to meet the conformance inspection, the lot represented by the sample shall be rejected.

4.6 Test procedures. Materials for all test procedures shall be manually mixed in accordance with the procedures described in 4.6.1 for the hardening test.

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4.6.1 Time for hardening. Specimens 9 by 4½ by 2½ inches shall be prepared according to ASTM C862 (except curing temperature) using the ball in-hand consistency of ASTM C860. After 24 hours curing, specimens shall be removed from the mold and the hardening characteristic shall be observed.

4.6.2 Modulus of rupture. Specimens shall be prepared in accordance with 4.6.1. Four specimens shall be tested for modulus of rupture after each of the following treatments:

a. Cured covered for 24 hours (±1 hour).

b. Cured, oven dried for 18 hours (overnight), and heat soaked at 1500 °F for 5 hours.

Heating at 1500 °F shall be in accordance with ASTM C865, except that the rate of temperature increase to 1500 °F may approximate that for group 16 of ASTM C210. Modulus of rupture of specimens at room temperature shall be determined in accordance with ASTM C133 test procedure (except specimen preparation).

4.6.3 Permanent linear change. Four specimens shall be prepared, cured, and dried as described in 4.6.1. The length of each specimen shall be measured to the nearest 0.01 inch after drying and again after heating at 2550 °F for 5 hours. Heating at 2550 °F shall be in accordance with ASTM C865, except that the heating schedule may be that shown for group 26 of ASTM C210. The percentage linear change, based on the dry length, shall be calculated for each specimen, and the average of the four shall be determined.

4.6.4 Bulk density. The bulk density test procedure uses acceptable specimens resulting from the hardening test. The test brick specimens, prepared as specified in 4.6.1 for the hardening test, shall be returned to the constant humidity chamber for an additional 24 hours of curing. The test bricks shall then be removed and oven dried for 24 hours at a temperature of 220-230 °F. Then the test bricks shall be measured. The test bricks shall be subjected to a 5-hour heat soak at the temperature shown for the type of castable mix being tested in accordance with table I, and then allowed to cool in the closed furnace to room temperature. The test bricks shall then be weighed. The density shall be computed in accordance with ASTM C134 from the oven dry volume and the weight after firing at the heat soak temperature.

4.6.5 Inclusion of drying agent. The packaging and internal process documentation shall be viewed to determine if polymeric fibers of the correct type and quantity were included in the castable prior to mixing, that the packaging reflects that the castable contains a drying agent, and that manufacturer's published drying procedures for the brand name supplied are correct for the included drying agent.

4.7 Toxicity and prohibited materials.

4.7.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.10.1, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the castable mix using data provided by the manufacturer/distributor to the NMCPHC (see 3.10.1 and 6.6).

4.7.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.10.2 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This castable mix will primarily be used in areas where standard brick shapes cannot be used. Type I, 3000 °F conventional castable mix, is used in shock-hardened burner front walls of naval boilers as a replacement for burner tile and adjacent high alumina firebrick and for general furnace repairs. Type II insulating castable mix is used in special applications around naval boiler superheater access doors where a high-temperature insulating castable is necessary. Type III lightweight insulating castable mix is used as a replacement for MIL-B-16008 (ASTM F1312) insulating brick where it is desirable to reduce insulating brick cutting around firebrick anchors. Types II and III insulating castable mix are not intended for direct exposure to furnace gasses. Existing stock of type I conventional castable mix manufactured to MIL-C-717F dated 12 November 1987 is acceptable for use until depleted. These castable mixes have been evaluated and tested for naval service conditions required by MIL-B-18381D that include high shock, rapid light-off and frequent light-off and securing cycles. These conditions are not considerations for all commercial products.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of refractory castable mix required (type I, II, or III) (see 1.2).

c. When first article inspection is required (see 3.1).

d. Requirements for an HHA (see 3.10.1 and 4.7.1).

e. Requirement for prohibited materials verification (see 3.10.2 and 4.7.2).

f. Unit of sample requirements if other than specified (see 4.4).

g. Requirement for test values and description of failures (see table V).

h. Packaging requirements (see 5.1).

i. Requirements for certificates of compliance (see 6.7 and 6.8).

j. When Safety Data Sheets (SDSs) submission and distribution is required (see 6.9).

k. Level of packing required including method of waterproofing pallets if required. This includes special packaging to extend the shelf-life for refractory castable mix to be stocked by the Government (see 6.11 and 6.12).

l. When palletization is not required (see 6.12).

m. Special marking requirements, if any (see 6.13).

6.3 First article. When a first article inspection is required (see 6.2), the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. A waiver can be recommended based on the contract award being less than 3 years from the date of previously acceptable first article testing for a particular castable mix brand name, and certification from the manufacturer that there has been no change in manufacturing location, fabrication process, composition, or source of basic heat-resistant raw materials since the earlier test, and no discontinuation of production for an extended period of time. The contracting officer can permit a waiver beyond the 3-year interval when there has been no discontinuation of production and certification from the manufacturer indicates that there has been no change in manufacturing location, fabrication process, composition, or source of basic heat-resistant raw materials since the earlier test.

6.4 Supersession data. This specification supersedes MIL-C-717F, dated 12 November 1987, and MIL-C-19794B, dated 11 December 1962.

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6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a material quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27,DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil.

6.5.1 Binding cement. In addition to packaging selected to extend shelf-life (see 6.10), suppliers should select brands that have the quantity and type of binding cement that will facilitate a longer shelf-life to meet the 3.8 requirement.

6.6 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.7 Certificates of compliance. The certificates of compliance (see 6.2) should include the results of verification testing for each requirement specified in section 3 and 6.2, as applicable. This will include the found test results, the specified requirements, and the conclusion that each result passed. For example, the modulus of rupture results for type I castable mix would be reported as 600 lb/in^2 (if that is the found test result) with the requirement of a 500 lb/in^2 (minimum) shown and the conclusion of "Passed" indicated. A copy of each certificate in addition to other reporting requirements of the contracts should be provided to Commanding Officer, NSWC Philadelphia, Attn: Code 412, 5001 South Broad St., Philadelphia, PA 19112.

6.8 Attached certification. Certificates of compliance (see 6.7) should be included with each shipment of castable mix. Certificates should indicate successful completion of the individual tests of the verification inspections as described in section 4, or state when previous testing was performed that was the basis for any waived testing inspections.

6.9 Safety data sheets (SDSs). When required, contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 will be in the contract.

6.10 Government packaging, preservation and marking instructions. The packaging instructions specified herein apply only for direct Government acquisitions when specified in 6.2.

6.11 Preservation and packaging. Preservation is considered applicable where extending castable mix shelf-life is desirable to accommodate storage and delay issues that may exist for the end user. Preservation for this purpose should consist of low Moisture Vapor Transmission Rate (MVTR) barrier materials with use of desiccant. The method described in 6.11.3.2 employs protective packaging to prevent physical damage to the MVTR barrier bags. When specified (see 6.2), special packaging requirements of level A and B (see 6.11.3) to extend shelf-life should be used.

6.11.1 Minimal packaging. Minimal packaging is intended for immediate or "just in time" delivery to the repair activity or where use within 6 months is anticipated and indoor, climate controlled, storage conditions can be provided. Applicable ASTM D3951 requirements for packaging, packing, and utilization apply.

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6.11.2 Commercial packaging. Currently most manufacturers will provide the product in paper reinforced shipping sacks with multi-wall moisture barrier protection (polyethylene lining). The following methods are preferable to extend shelf-life: when a manufacturer can provide thicker (greater than 10 mils) heat sealed polyethylene sacks or when a manufacturer can fill 50-pound High Density Polyethylene (HDPE) pails with castable mix. NOTE: Pails 14½ inches (368.30 millimeters) in length (not to be exceeded) and 11⅞ inches (301.62 millimeters) in diameter should be used. Unless the pail is being directly filled at the point of manufacture, this method should not be permitted, since filling at a remote location will create a second exposure of the contents to moisture. See 6.11.3.1 for other characteristics of the pails.

6.11.3 Level A and B packaging and preservation. Level A or B packaging should be as follows when specified (see 6.2). Refractory castable mix should be furnished in unit quantities of 50 pounds (22.68 kilograms), 55 pounds (24.95 kilograms), or 60 pounds (27.22 kilograms) in accordance with the manufacturer's practice. The Original Equipment Manufacturer (OEM) commercial packaging (see 6.11.2) is then enclosed in plastic pails with covers conforming to 6.11.3.1 or heat sealed MIL-PRF-117 bags in accordance with 6.11.3.2. The date of manufacture should be transferred from the OEM bag to the outer container. Packing for MIL-STD-2073-1 level B should be as specified for level A except that boxes referred to in 6.11.3.2 should conform to ASTM D4727/D4727M class domestic (D).

6.11.3.1 HDPE pails.

a. The pail should be molded polyethylene with a minimum 90-mil thickness.

b. Should be stackable and self-supporting, reinforced to prevent ovaling.

c. Should be sized to accept the OEM's sealed bag in its entirety. This must include the necessary oversizing so that the bag can be inserted without distortion or damage. Pails may be tapered to permit mating.

d. Should include a metal bail with handgrip. Bails should be minimum 7-gauge wire (standard wire gauge system).

e. Bail ears should be integrally molded near the top of the pail.

f. Pail covers should be minimum 16-gauge steel or minimum 90-mil polyethylene. Covers should be gasketed to assure a watertight seal. Metal covers should not affect nor be affected by the castable mix.

g. Should contain bagged, activated, desiccant in accordance with MIL-D-3464 type I. The desiccant should be added to each pail after insertion of the OEM bagged refractory. Forty-eight desiccant units (24 units on each side of the castable mix bag) should be used. This will increase the weight of the pail by about 3.7 pounds. A humidity indicator card conforming to MIL-I-8835 (three spot MS20003-2 design) should be attached to the underside of the cover of each pail.

6.11.3.2 MIL-DTL-117 bags. As an alternative to HDPE pails, MIL-STD-2073-1, method 51 can be used. The OEM castable bag should be enclosed in a MIL-DTL-117, type I, class E, style 1 bag (using MIL-PRF-131, class 1 Low MTVR barrier materials). The material should be super heavy duty with Biax nylon/PE/cross-lam PE/PE/foil/heavy duty coex structure of 10-mil minimum caliber. Two eight-unit desiccant packs, MIL-D-3464, type I, should be positioned at each end of the OEM bag, end to end, with fiberboard divider (see below) inserted outside of the desiccant. The fiberboard should be taped to the OEM castable bag to hold the desiccant in place for insertion in the outer bag. The outer bag should be close fitted, but still allow the refractory castable mix bag to be inserted without damage to either bag. A humidity indicator card (see 6.11.3.1.g) should be inserted in the MIL-DTL-117 barrier bag window, and excess air should be manually squeezed out of the outer bag before sealing. The ASTM D4727/D4727M, type CF (corrugated fiberboard) single wall rectangular divider should be the size of the OEM bag sealed ends with corners removed. A fiberboard box, as follows, should be required to complete the unit container:

a. Fiberboard – ASTM D4727/D4727M, variety SW (single wall), type CF, class WWVR, grade V3c

b. Box – ASTM D5118/D5118M, style FOL/0203

c. Closure method – ASTM D1974/D1974M, method 4B1

d. Reinforcement method – ASTM D1974/D1974M, method 2B

The primary cushioning specified in the contract should be placed between the outside bag and the inside of the carton. The box size should be selected to minimize the use of cushioning material.

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6.11.3.3 Special packaging inspection. When special packaging for overseas shipment or for extension of shelf-life is specified in the contract (see 6.2), the packaging should be visually examined for workmanship, including for defects in sealing, absence of physical damage, failure of desiccant, moisture damage of product, correct type of barrier packaging, transference of manufacturing date to outer container, and general compliance with required delivery preparations. The package markings should be visually examined for conformance with requirements and verified they are consistent with the castable type being supplied. Where desiccant and humidity indication cards are required with the use of outer vapor barrier packaging, the type and quantity and viability should be verified.

6.12 Palletization. Unless otherwise specified (see 6.2), castable mix should be palletized in accordance with MIL-STD-147, load type IV for pails and load type XV for boxed sacks. Stretch wrap or shrink film may be used in lieu of strapping. For palletized boxed sacks 1 inch by 4 inches, wood slats should be placed with a nominal 4-inch dimension arranged vertically along the perimeter of the pallet to protect the layer of boxes from being damaged by forklift equipment. Pallet loads should not exceed 2500 pounds (1134 kilograms). Boxes should not be stacked more than three high and pails should not be stacked more than two high.

6.13 Marking. In addition to any special marking required (see 6.2), pails, boxed sacks, and palletized unit loads should be marked in accordance with MIL-STD-129 for levels A and B and in accordance with ASTM D3951 for commercial packaging. Containers should include the date (month and year) the castable mix was packed by the OEM on the outside of the pail or box. Instructions for mixing and application should be marked on each container. These instructions should address the curing time and the presence and type of polymeric de-watering additives. The date of manufacture must be marked on the container (see 6.11.3).

6.14 Subject term (key word) listing.

Density

Heat-resistant

Linear change

Preheat

Rupture

Shelf-life

Spalling

6.15 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 9350-2018-001)

Review activities:
Army – MR
DLA – CQ

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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